CHINENYEZE FRANCIS ONYEMA

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PROFESSIONAL SUMMARY

Board-certified Prosthetist and Orthotist with over 4 years of clinical experience delivering patient-cantered care across military reference hospitals, private practices, and digital fabrication environments. Expert in patient assessment, biomechanical gait analysis, custom orthotic/prosthetic design, and precise physical and digital fabrication. Demonstrated success in mentoring undergraduate students, collaborating with multidisciplinary healthcare teams, and volunteering as an advocate for amputee care. Dedicated to restoring patient mobility and improving quality of life through tailored, evidence-based interventions.

EDUCATION

B. Tech Prosthetics and Orthotics 2014-2019

Federal University of Technology Owerri. HTTP://futo.edu.ng

RESEARCH INTEREST

Rehabilitation engineering & clinical gait analysis Foot and ankle biomechanics Orthotics, prosthetics, exoskeletons, footwear.

RELEVANT COURSEWORK

Anatomy

Physiology

Advanced Biomechanics

Dynamics and statics

Kinesiology

Neuroscience

RESEARCH EXPERIENCE

Federal University of Technology Owerri.

Thesis: The Effects of gasoline fumes on the Bone Histo-architecture of albino wistar rats

- This study analyses how gasoline fumes affect the bone histology architecture of Wistar albino rats over 12 weeks of exposure.

Advisor: Dr CU Okeke

TEACHING AND MENTORING EXPERIENCE

Undergraduate students (Dept of prosthetics and Orthotics) Shehu Idris College of Health Sciences and Technology, Kaduna

Amina Bashir, Patience Pyongyang, Kingsley Stephens, Aisha Umar, Saminu Abdul mentored and taught undergraduate

students on

- Biomechanics of prosthetics and orthotics
- prosthetic designs and technology
- Socket technologies
- clinical gait analysis
- Medical ethics and practice

COMMUNITY OUTREACH AND SERVICE

The IREDE Foundation, August 2016-present, volunteer writer and lead advocate:

I write articles on the overall well-being of amputees and on ways they can live full lives. The IREDE Foundation empowers, encourages, educates, and provides prosthetic limbs for children aged 0–18 in Nigeria. We seek to ensure no child should be out of school because of their limb difference.

- Cultivated social media following across Twitter with more than 1,000 followers.
- I wrote and edited high-quality content and visually impactful programs under deadline pressure using exciting, captivating, and authentic approaches.
- Completed thorough research on the assigned topics.
- Maintained a current understanding of industry and genre trends, enhancing writing and marketing success. • Consulted editors to shape stories and eliminate any errors.
- Brainstormed with the type department to determine new topics and stories.

LEAD ADVOCATE:

Out on a limb 2022 (April) Theme: Iredians – the legends of hope Rehabica

July 2020-present Volunteer member:

In June 2020, a team of medical rehabilitation professionals set up a community of African medical rehabilitation professionals called "REHABICA".

As a volunteer member, I am responsible for the growth of the community, providing updates on continuing education on prosthetics and orthotics, new innovations on prosthetics and orthotics, and meetings on updates on personal professional development.

Rehab Edge, Sept 2019-present Volunteer writer:

- Utilized exceptional ting, writing, editing and proofreading skills to produce engaging and error-free content • Effectively address the target audience and the purpose of communication with the proper tone, sentence structure, and word count, ensuring updated trends in the rehabilitation space.

CLINICAL EXPERIENCE

INSTALIMB JAPAN NOVEMBER 2025 TILL DATE:

Prosthetist and orthotist:

Technical and Clinical Duties

Clinical Roles

- Conduct Patient assessment and P&O prescription support.
- Perform Anthropometric measurements; 3D scanning of residual limbs/body segments. • 3D CAD-based design of sockets and orthoses; alignment and trial fittings.
- Evaluate Post-fitting gait/mobility and iterative adjustments for better outcomes.
- Patient education on use, care, and maintenance.

• Maintain accurate clinical documentation of patient data, components, and adjustment history.

Fabrication Tasks

• Interpret the digital design specification, the required tools, and materials.

• Process printer files; operate and calibrate 3D printers; manage consumables.

• Perform routine maintenance and basic repairs; adhered to safety practices.

• Support with production scheduling and inventory control where needed.

IT Systems Support

• Supported rollout and training for "Instalimb ONE" (customer & production management). •

Collect user feedback and implementation gaps, report to HQ for improvements.

Clinical Engagement & Technical Supervision

• Supervised patient evaluation, fitting, and follow-up at other partner sites.

• Guide safe clinical practice and ensure quality/traceability in the digital workflow.

• Gathered and analyzed patient-reported outcomes and feedback for service improvement

Training & Technical Support for Partner Sites

• Deliver structured training on digital P&O workflows (3D scanning → CAD design → 3D printing → assembly → fitting → follow-up).

• Used standard training manuals; conduct skills assessments and refresher sessions.

• Provided ongoing troubleshooting and mentorship (in person and remote).

Product & Process Localization

• Recommend adjustments to device specifications based on local anthropometrics, climate, activity levels, and component availability.

• Conducted local materials or process improvement studies with the global R&D team for better improvement outcomes

• Prototype and evaluate localized configurations with clear documentation

Documentation & reporting

• Maintained detailed records, including training logs, clinical supervision notes, non-conformance reports, and corrective actions.

• Consolidated manufacturing documentation from partner sites.

• Prepare briefings/reports for internal/external stakeholders

SUN-TECH PROSTHETICS AND ORTHOTICS, ABUJA, JANUARY 2023- OCTOBER 2025

Certified prosthetist and orthotist:

• Instructed patients on use and care of fitted orthoses to enhance performance and longevity. • Designed and constructed orthoses to address wide range of conditions, including spinal deformities, scoliosis and other musculoskeletal issues.

• Diagnosed conditions and developed treatment plan to restore movement, reduce pain and improve mobility. •

Participated in research and development of new orthotic devices and techniques.

• Developed customized orthotic devices to meet unique needs of each patient.

• Used varied materials to fabricate and fit orthoses for patients.

• Conducted patient consultations to gather information regarding medical history and orthotic needs.

• Monitored patient progress to obtain optimal outcomes.

• Provided orthotic treatments to patients with mobility issues to improve quality of life.

• Collaborated with physicians and other healthcare professionals to provide best outcomes for patients. • Evaluated patients to determine individual orthotic needs.

• performed biomechanical analysis on prosthetic and orthotic device

• Performed maintenance and repair of orthotic devices to avoid discomfort and potential damage. • Provided support and guidance to patients and caregivers during orthotic treatment process. • kept up-to-date with new materials and procedures through journaling, electronic media, and correspondence with peers.

• Maintained accurate patient records with measurements, drawings, and other relevant data for future reference. • Read prescriptions or specifications to determine the type of product or device needing fabrication. • Provided follow-up care to facilitate optimal patient outcomes.

• Developed treatment plans tailored to individual patient needs.

44 NIGERIAN ARMY REFERENCE HOSPITAL, KADUNA, JULY 2021-AUGUST 2022.

Certified Prosthetist and Orthotist:

• Evaluate and interview the patient to determine their needs.

• Take measurements or impression of the body part of a patient that will be fitted with a brace or artificial limb. • Assist the Prosthetist in the design and fabrication of orthotic and prosthetic device instruct patient on proper use and care of their devices.

• Document care in patient's record.

• Performed static and dynamic alignment

• Meet individual patient needs by discussing optimal treatment plan benefits, risks and time involved with each choice.

• Mentored prosthetics and orthotics students during their clinical posting on biomechanical impacts on prosthetics and orthotics, prosthetic designs and technology and orthotic interventions for patients with neuromuscular disorders

FIX AND FIT PROSTHETICS AND ORTHOTICS, ABUJA OCTOBER 2019- DECEMBER 2019

Student intern:

• learnt the clinical and technical aspects of prosthetics and orthotics ranging from patient assessment and evaluation, fabrication, alignment and fitting of the patient.

• Assisted with Repairs or updated Prosthetic and orthotic devices to support patients' needs • Assisted with the modification of plaster cast of area fitted with Prosthesis and orthoses, for use in device construction process.

NATIONAL HOSPITAL ABUJA OCTOBER 2018 - DECEMBER 2018

Student intern:

• learnt the basics of plaster of Paris cast application

• learnt the clinical aspect of Orthopaedics by participating in ward rounds and outpatient clinics • worked with the rehabilitation team to decide on the appropriate prosthetic and orthotic device for patients performed initial patient evaluation and formulate treatment plans

NCDC-NYSC COVID TEAM:

Member/Data Collection officer;

• Electronic data collection using the surveillance outbreak response.

• Management and analysis system (SORMAS) platform with buffer filling for the COVID- 19 Testing •

Collated data for over 3,000 corps members Sensitization for COVID-19

TECHNICAL SKILLS

- Python
- Microsoft Office (Word, Excel, PowerPoint) Powerbi
- Autodesk fusion360 Autodesk inventor
- 3D modelling
- 3D Scanning
- 3D printing
- 3D printer maintenance

SOFT SKILLS

- Good communication
- Strong attention to detail
- Excellent interpersonal skills
- Manuel dexterity
- Ability to work in a team setting effectively.

SPECIALTY CLINICAL SKILLS AND TRAINING

INSTALIMB DIGITAL FABRICATION PROGRAM

- Practitioner level certification in 3D technology
- Level 1 and 2,3D printer maintenance certification

BAYLORS COLLEGE OF MEDICINE;

Department of prosthetics and orthotics, Orthotics and prosthetics continuing professional development sessions

- Physical examination in orthotics and prosthetics
- Technical skills in orthotics and prosthetics
- Practice management and professionalism
- Lower limb orthotics
- Cranial management
- Upper limb orthotics and prosthetics
- Spinal and scoliosis management

Cascade Dafo Institute:

- Using Dafo in treatment plans for patients with cerebral palsy
- Toe walking intensive
- Assessment of Dafo
- Brace planning for crouched gait

Allard USA:

- Applied biomechanics in the management of partial foot amputation
- Prosthetic management of partial foot amputation
- Functional biomechanics improve heel-to-hip linkage to restore gait

Thuasne USA

- Custom carbon Afos and the influence of the plantar flexors
- Burns and Scar Management – Insights from a Clinical Practice in Germany
- Effects of Dynamic Knee Braces on OA: Insights from MAI Motion Analysis

• Re-thinking Early Orthotic Intervention in Post-Stroke Patients
• Introducing Agilik™ - A new, smart powered orthosis

Becker Orthopaedic

• Friction management for neuropathic foot problems.

• Biomechanics of partial foot amputation

PROFESSIONAL SOCIETIES

• Nigerian prosthetic, orthotic and orthopaedic technology society. September 2014
• Nigerian red cross society June 2018
• Medical rehabilitation and therapy board of Nigeria Feb 2021
• Black biomechanics association 2022

PROFESSIONAL CERTIFICATIONS

Medical rehabilitation and therapy board of Nigeria: Certified Prosthetist and Orthotist February 2022 Nigeria red cross society:

First Aid /CPR August 2019

LANGUAGE SKILLS

English: Native language, Distinguished levels in listening, Speaking, Reading and writing;

Swedish: beginners

Dutch: beginners

PUBLICATIONS

“Tips for amplifying your success as a lower limb prosthetic user”, Irede medium stories, October 2019 “The

K-Levels for Lower Limb Amputees”, Irede medium stories, September 2019

CONFERENCES

• 1st scientific conference/ AGM and workshop of Nigerian Prosthetic, Orthotic and Orthopedic Technology Society Owerri 2016
• 2nd International Conference of Medical rehabilitation professionals (ICMRP) Abuja - Strategic initiatives for sustainable medical rehabilitation services in Nigeria and sub- sahara Africa.
• 3rd annual scientific conference and annual general meeting -implementing government policies on disability management in Nigeria.2024